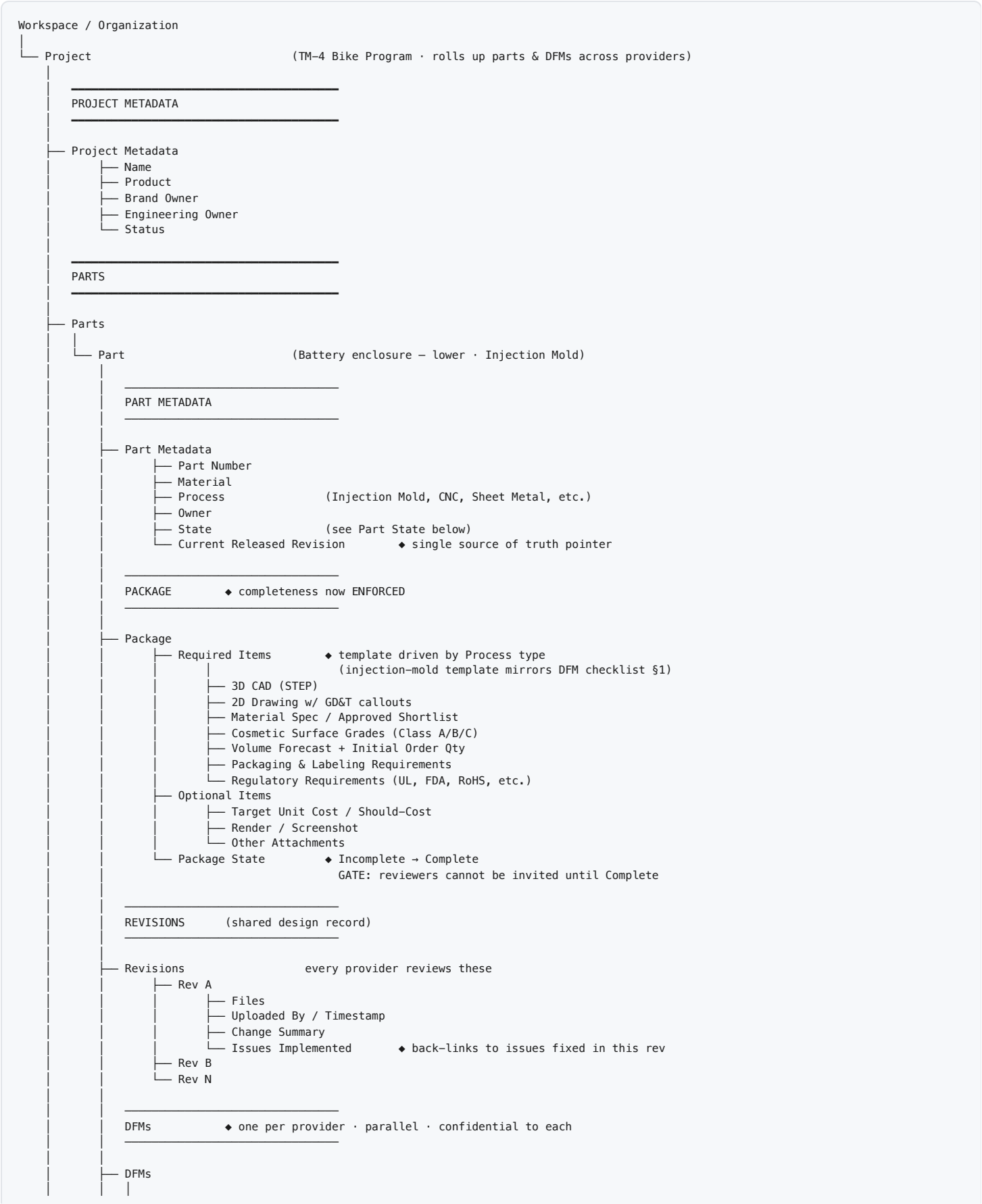


Chorus v0 — Full Hierarchy

DFM-Only, Multi-Provider · Changes from previous version marked with ♦



- DFM (Hsinchu Precision · CM · scoped magic link)
 - DFM Metadata
 - Provider (→ External Reviewer)
 - Role (CM, Supplier, Fabricator, Tooling)
 - Confidentiality Boundary ♦ providers never see each other's DFMs
 - Current Revision Under Review ♦ per-DFM pointer – Provider 1 can be on Rev B while Provider 2 is on Rev A

ISSUES ← the atomic unit of work

- Issues
 - Issue (#12 Insufficient draft for ejection)
 - Title / Description
 - Type ♦ NEW axis – findings ≠ proposals
 - Show-stopper (not moldable as designed)
 - Finding (problem needing a design change)
 - Proposal (parting line, gate location, shrink est.)
 - Info (no action required)
 - Category
 - Geometry / Tolerance / Material
 - Tooling / Assembly / Process / Cost-Yield Risk
 - Severity (Critical / High / Medium / Low)
 - Created On Revision / Created By
 - Recommendation (provider's proposed fix)
 - Cost / Yield Impact ♦ optional \$ field – future quote hook
 - Brand Decision
 - Accepted / Rejected / Needs Clarification
 - Rationale ♦ REQUIRED on reject (checklist §3)
 - Implementation
 - Not Started / In Progress
 - Implemented in Rev X ♦ explicit rev link
 - Validation
 - Owner = reviewer who raised the issue ♦ supplier re-confirms
 - Compares: created-on rev vs implemented-in rev
 - Pending / Validated / Validation Failed (→ reopens)
 - Derived Issue State ♦ Open → Dispositioned → Implemented → Validated → Closed
 - Comments / Attachments
 - Audit Trail
 - DFM State ♦ Invited → In Review → Feedback Submitted → Awaiting Validation → Complete

ISSUE GROUPS ♦ NEW · brand-side only · cross-DFM rollup

- Issue Groups
 - Group ("Wall thickness – boss area" · 3 providers flagged)
 - Linked Issues (one per provider DFM)
 - Conflict Flag ♦ providers gave contradictory recommendations
 - Unified Decision ♦ disposition once → cascades to linked issues
 - Confidentiality preserved: providers see only their own issue

SIGN-OFFS ♦ NEW · joint alignments are first-class, not issues

- Sign-offs
 - Sign-off (Gate location · Brand + Hsinchu Precision)
 - Topic (Parting Line / Gate Location / Material Lock / Tooling Ownership / custom)
 - Parties (brand owner + provider, both must confirm)
 - State Proposed → Aligned → Signed
 - Record (what was agreed, written rationale)
 - Audit Trail

DFM APPROVAL ♦ NEW · the gate that matters (checklist §6 – cut steel)

- DFM Approval
 - Entry Criteria ALL open issues dispositioned + sign-offs signed
 - Approved By / Timestamp
 - Approved Revision (frozen – the rev steel gets cut against)
 - P0 / Tooling Reference (optional external ref)

PART STATE ♦ ends at approval, not "complete"

Part State

- Draft
- Package Complete ♦ new gate state
- DFM Active
- Awaiting Validation
- DFM Approved ♦ replaces "DFM Complete" – cleared to cut steel

ACTIVITY

Activity

- Revision Uploaded / Issue Created / Issue Updated
- Decision Recorded / Validation Requested / Issue Closed
- Sign-off Recorded ♦
- DFM Approved ♦

EXTERNAL REVIEWERS

External Reviewers

- Company
- Role (CM, Supplier, Fabricator, Tooling, etc.)
- NDA Status ♦ gate before any files are visible
- Access Scope (which parts → which DFMs · nothing else)
- Magic Link
- Expiry / Revoke ♦
- File Watermarking Policy ♦ CAD/drawings stamped per recipient
- Reviewer Activity Log ♦ who viewed/downloaded what, when

PROJECT ACTIVITY FEED

Project Activity Feed

- Reviewer Invited / Part Added / Revision Uploaded
- Issue Opened / Decision Recorded / Validation Requested / Issue Closed
- Sign-off Recorded ♦
- DFM Approved ♦

PROJECT STATE

Project State

- Setup
- DFM Active
- DFM Approved ♦

vNEXT HOOKS ♦ out of scope, modeled as stubs

vNext Hooks

- Tooling Quote (checklist §4 – attaches to DFM Approval)
- Mold Flow / Simulation (checklist §5 – attaches to Revision)
- T1 / FAI / PPAP (checklist §7 – attaches to Approved Revision)

The six fixes, in one line each

1. **Package gate** — reviewers can't be invited until the OEM package meets the process-type checklist; bad inputs were the #1 cause of useless DFM feedback.
2. **Issue Type axis** — show-stoppers, findings, proposals, and info notes flow through disposition differently; proposals were polluting the accept/reject queue.
3. **Per-DFM revision pointer** — providers review at their own pace; validation always compares the rev an issue was raised on vs the rev that fixed it.
4. **Issue Groups** — brand dispositions one cross-provider duplicate once, with a conflict flag when providers disagree; confidentiality boundary holds.
5. **Sign-offs** — parting line, gate location, material lock are joint agreements between named parties, not issues one side "accepts."

6. **DFM Approval gate** — the terminal state is the steel-cut decision (all issues dispositioned + sign-offs signed + frozen revision), the moment Chorus's value is realized and the natural anchor for pricing.